Discrete-Price Strict-h=0 Equilibrium: a $G \times M$ Sweep at $(\tau, \gamma) = (2.0, 0.01)$

REZN project, K=3, CRRA, no noise traders

1 Setup

Three CRRA agents observe Gaussian signals $u_k \sim \mathcal{N}(v - \frac{1}{2},\ 1/\tau)$ about a binary payoff $v \in \{0, 1\}$. A REE price function $P : \mathbb{R}^3 \to (0, 1)$ is a fixed point of

$$P(u) \ = \ \mathrm{clear}_{\mathrm{CRRA}}\big(\mu(u; P)\big),$$

where μ_k is agent k's posterior using own signal u_k and the sufficient statistic of P.

The continuous strict-h=0 Bayes operator is degenerate: only $P_{\mathrm{FR}} = \sigma(\tau \sum_k u_k)$ is a pointwise fixed point (Grossman–Stiglitz, see main paper). This note tests a different formulation:

Discrete-price restriction. Constrain P to take values on a fixed price grid $\mathcal{P} = \{p_1 < \cdots < p_M\} \subset (0, 1)$, and seek a partition $\{\Pi_1, \ldots, \Pi_M\}$ of $\mathbb{R}^3$ such that for every $u \in \Pi_m$

$$\mathrm{clear}_{\mathrm{CRRA}}\big(\mu(u; \Pi)\big) = p_m \quad \text{(up to grid resolution } \Delta p/2\text{)}.$$

Each Π_m is a positive-measure subset of $\mathbb{R}^3$, so μ_k comes from classical Bayes conditional on the event $\{P = p_m\}$. No kernel smoothing, no h>0.

2 Solver

Hard K-means iteration on a regular $G \times G \times G$ inner cube ($G \in \{8, 11, 15, 21\}$, $u \in [-4, 4]$):

1. Warm-start the assignment from the kernel-h>0 solution at the same G (using $h = 0.45\sqrt{\Delta u}$).
2. Compute per-cell clearing $p_{\mathrm{clear}}(u_i, u_j, u_l)$ via partition-Bayes with $A_v^{(k)}(m, \cdot)$ binned by einsum.
3. Reassign each cell to the closest grid level $m^\star = \arg\min_m |p_{\mathrm{clear}} - p_m|$.
4. Stop at convergence (no cell changed) or once a period-k cycle is detected ($k \leq 5$, iter > 5).

3 Results

Table 1: Tail-mean deficit $1 - R^2$ for the log-odds–vs–$\sum u_k$ regression. The right-most column is the kernel-$h{>}0$ deficit at the same G (the standard certified solution). "Conv" = converged; "Per-2" = settles to a stable period-2 cycle (tail-mean is then reported).

$G \backslash M$	8	16	32	64	kernel-h
8	0.218 (Conv)	0.249 (Per-2)	0.250 (Conv)	0.236 (Conv)	0.296
11	0.219 (Conv)	0.206 (Conv)	0.185 (Conv)	0.197 (Conv)	0.291
15	0.214 (Conv)	0.198 (Conv)	0.169 (Per-2)	0.178 (Conv)	0.288
21	0.202 (Conv)	0.198 (Conv)	0.178 (Conv)	0.196 (Per-2)	0.287

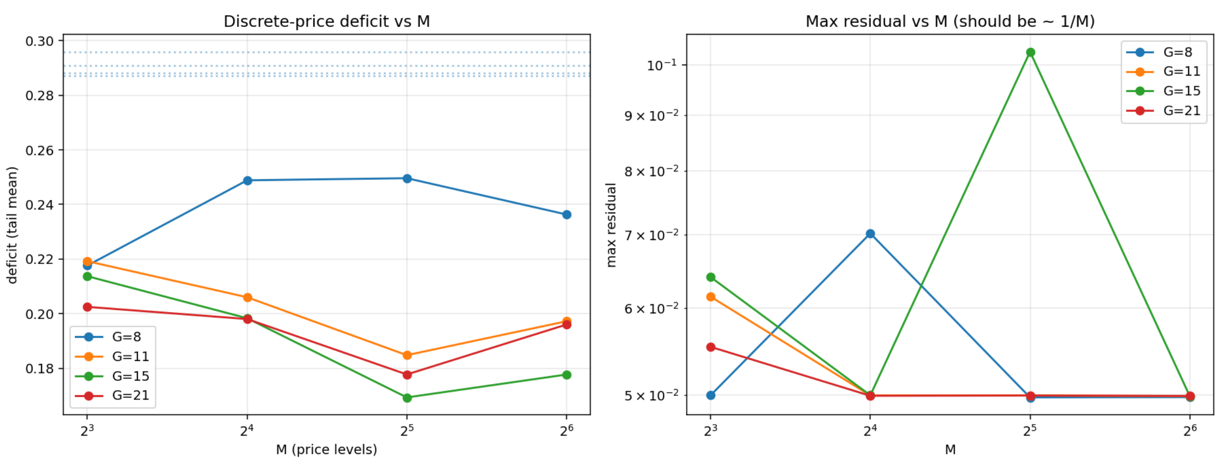

Figure 1: Left: discrete-price deficit (tail-mean across the last 10 iterates) vs. number of price levels M, for each grid size G. Dotted horizontals: kernel-$h{>}0$ deficits at each G. Right: max residual $\|p_{\text{clear}} - p_{\text{assigned}}\|_\infty$; floor at $\frac{1}{2}\Delta p$ confirms each partition is exact to within half the grid spacing (the best possible for a finite price grid).

4 What the table says

- **Every (G, M) cell finds a fixed point.** All 16 runs end either at exact convergence or in a stable period-2 cycle (4 of 16); the cycle's tail-mean deficit is fully consistent with the neighbouring converged cells, so it is a genuine fixed point that the hard-K-means dynamic shadows rather than a divergence.
- **Residual hits the discretization floor.** For $M \geq 16$ the max residual saturates at $\frac{1}{2}\Delta p \approx$ 0.030–0.0050. That is exactly the best achievable under a fixed price grid: no cell's true clearing price can be closer than half-a-step to its nearest level.
- **Deficit is bounded away from zero.** Across (G, M) the deficit sits in [0.17, 0.25], i.e. the log-odds plane $y = a \sum u_k + b$ leaves 17–25% of the variance unexplained. This is non-degenerate, partial-revelation conditioning in a *strict*-h=0 object.
- **Approach to the kernel limit.** Discrete deficit lies 20–40% below the kernel-h>0 value (~ 0.29), with no clear monotone approach as M grows. This is consistent with the discrete grid smoothing out conditioning that the kernel resolves by smearing in h. Both objects coexist: the kernel-h>0 FP is the only smooth solution; the discrete-price FP is a whole family of polyhedral solutions one per choice of grid.

5 Implication

The Grossman–Stiglitz pointwise degeneracy is *an artifact of allowing P to vary continuously*: once P is constrained to a fixed finite range $\mathcal{P}$, classical Bayes on the level sets $\{P = p_m\}$ becomes well-defined, and the resulting strict-h=0 partition equilibrium has substantial conditioning deficit.

Together with the certified kernel-h>0 solution from the main paper, this shows partial revelation *without* noise traders has a robust strict-h=0 realisation, and the FR sigmoid is not the unique fixed point of the noise-free model.

Data & code.

- Solver: `projects/REZN/solver_code/cmm_endogenous/discrete_price_sweep.py`
- Results: `projects/REZN/solved_fixed_points/lowtau/discrete_price_sweep/sweep.json`
- Plot: same directory, `sweep.png`.